Week 7 Homework

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In order to receive full credit problems and solutions must be clearly written and presented neatly. In writing up your solution, you should state the problem as explaining the solution. Merely giving answers to exercises will is not enough to receive credit. Your write up must show work and convince me that you understand the material. If you need additional time to complete this assignment, please email me. Starred () proof problems are eligible for inclusion in the proof portfolio assignment.

Exercises

Exercise

Find a recurrence relation for the number, b_n of length n binary strings with no two consecutive ones (No 11 in the string). What are the initial conditions? What is the closed form solution to this recurrence relationship?

Exercise

Solve the following homogeneous recurrence relations:

1. $a_n = 5a_{n-1} - 6a_{n-2}$ with $a_1 = -1$ and $a_2 = 1$
2. $b_n = 6b_{n-1} - 9b_{n-2}$ with $b_1 = 1$ and $b_2 = 9$

Exercise

Solve the following homogeneous recurrence relations:

1. $a_n = 6a_{n-1} - 8a_{n-2}$; $a_0 = 1$, $a_1 = 0$
2. $a_n = 2a_{n-1} + 8a_{n-2}$; $a_0 = 4$, $a_1 = 10$

Exercise

Consider tiling a $2 \times n$ board with 2×1 and 2×2 tiles.

1. Let a_n count the number of tilings of where the 2×1 can only be positioned vertically . Find a recursive relationship for a_n. Find initial conditions for a_n.
2. Let b_n count the number of tilings of where the 2×1 can only be positioned horizontally. Find a recursive relationship for b_n and find initial conditions for b_n.
3. Let c_n count the number of tilings of where the 2×1 can be positioned horizontally or vertically. Find a recursive relationship for c_n and find initial conditions for c_n.

Exercise

Let s_n be the number of length-n bit strings with no three consecutive 0s. Find s_1, s_2, s_3 and find a recursive definition of this sequence.

Exercise

Find a closed form solution to recurrence relation $x_n = n$ for $0 \leq n < m$ and $x_n = x_{n-m} + 1$ for $n \geq m$.

Exercise

What is dynamic programing? Describe the basic algorithm and explain its connection to recurrence relations.

Exercise

What is the Josephus number? Define it and give its historical origins. What is its connection to recurrence relations?

Exercise

A string that contains only 0's,1's, and 2's is called a **ternary string**. Find a recurrence relation for the number of ternary strings that contain two consecutive 0s? What are the initial conditions? Explain your answer. What is the recursion and initial conditions for ternary strings that contain 000?

Exercise

Each day Angela eats lunch at a deli, ordering one of the following: chicken salad, a tuna sandwich, or a turkey wrap. Find a recurrence relation for the number of ways for her to order lunch for the "n" days if she never orders chicken salad three days in a row.

Exercise

Find the number of non-negative integer solutions to $x + y + z + w \leq 6$. Generalize this to find the number of non-negative integer solutions to $x + y + z + w \leq k$.

Exercise

Let $F_{n+2} = F_{n+1} + F_n$ with $F_0 = 0$ and $F_1 = 1$, $L_{n+2} = L_{n+1} + L_n$ with $L_0 = 2$ and $L_1 = 1$. Find a relationship between L_n, F_{n-1}, and F_{n+1}.

Exercise

Find the first 5 terms of the Taylor expansion at $x = 0$ of $\sqrt{1-4x}$. By examining the pattern of coefficients find a_n where $\sqrt{1-4x} = \sum_{n=0}^{\infty} a_n x^n$.

Exercise

Use the Auxiliary Equation method to find a closes form solution to the following recurrence relations:

1. $c_n = 3c_{n-1} - 3c_{n-2} + c_{n-3}$ with $c_0 = 1$, $c_1 = 2$, and $c_2 = 4$
2. $x_n = 5x_{n-1} - 4x_{n-2} - 6x_{n-3}$ with $x_0 = 0$, $x_1 = 0$, and $x_2 = 1$

Proofs

Proof

Prove

$$\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n (n+1)!} 4^n = \frac{1}{n+1} \binom{2n}{n}.$$

Proof

Prove $\binom{3n}{3} = 3\binom{n}{3} + 6n\binom{n}{2} + n^2\binom{n}{1}$.

Proof

Using induction, prove that for the sequence defined by $a_1 = 0$, $a_2 = 1/2$, and $a_{n+2} = \frac{1}{2}(a_n + a_{n+1})$ for $n \geq 1$ that

$$a_n = \frac{1}{3}\left(1 - \left(-\frac{1}{2}\right)^{n-1}\right)$$

for every $n \geq 1$.

Proof

Consider the sequence defined by $a_n = \frac{1}{2}a_{n-1} + c$ where $a_1 = c$. Prove that $a_n = c(2 - \frac{1}{2^{n-1}})$ using mathematical induction.

Proof

Prove

$$\sum_{k\geq 0}^{n} k\binom{n}{k} = n2^{n-1}$$

Proof

Let $T_n = \frac{n(n+1)}{2}$ be the triangular numbers and $P_n = \frac{3n^2-n}{2}$ be the pentagonal numbers. Using the direct proof method, prove $T_{5n-2} = T_{2n-2} + 6P_n + 3T_{n-1}$ for all $n \geq 1$.

Proof

Prove

$$\sum_{i=0}^{k}(-1)^i\binom{n}{i} = (-1)^k\binom{n-1}{k}$$

Proof

Prove

$$\sum_{k\geq 0} k\binom{m}{k}\binom{n}{k} = n\binom{m+n-1}{n}$$

Hint: See Example 1.10 and 1.11 on page 9 of the course text.

Proof

Consider the following modification of the Towers of Hanoi problem with three posts labeled A, B, and C with the added two rules 1) A disk may be moved only from a post and adjacent post, and 2) the disks must be moved from post A to post C. Justify a recursive description of this problem, and prove by induction that the number of moves need to move n disks is $3^n - 1$.

Proof

Let $F_{n+2} = F_{n+1}+F_n$ with $F_0 = 0$ and $F_1 = 1$, the Fibbonacci numbers, and $L_{n+2} = L_{n+1}+L_n$ with $L_0 = 2$ and $L_1 = 1$, the Lucas numbers. Prove

$$\sum_{k=0}^{n} 2^k L_k = 2^{n+1}F_{n+1}$$

for all $n \geq 0$.

Proof

Let $F_{n+2} = F_{n+1}+F_n$ with $F_0 = 0$ and $F_1 = 1$, the Fibbonacci numbers, and $L_{n+2} = L_{n+1}+L_n$ with $L_0 = 2$ and $L_1 = 1$, the Lucas numbers. Prove

$$\sum_{k=0}^{n}(-1)^k 2^{n-k} L_{k+1} = (-1)^n F_{n+1}$$

for all $n \geq 0$.

Proof

Define $x_1 = 1$ and $x_2 = 2$ and $x_{n+2} = x_{n+1} + x_n$ for $n \geq 1$. Prove that $4^n x_n < 7^n$ for all positive integers n.

Proof

Prove $\binom{n^2}{2} = n\binom{n}{2} + n^2\binom{n}{2}$.